

NAME: VINAY KUMAR BATHULA

EMAIL: vinaybathula04@gmail.com

PHONE: (341)226-8902

SENIOR BIG DATA ENGINEER

PROFESSIONAL SUMMARY

- Over 10+ years of experience in designing, developing, and deploying data pipelines and big data analytics solutions across cloud platforms and distributed computing environments.
- Proficient in the **Hadoop** ecosystem including **HDFS**, **MapReduce**, **YARN**, **Hive**, **Pig**, **Sqoop**, **Flume**, **Impala**, and **Oozie** for efficient big data management and processing.
- Expert in developing **scalable** and high-performance data pipelines using **Apache Spark (Scala/Python)**, **Spark Streaming**, **Kafka**, and **Apache Flink**.
- Experienced in implementing data governance and access controls using **Unity Catalog** and building scalable data solutions on **Databricks**.
- Developed scalable ETL pipelines using **Databricks notebooks** with **PySpark** for large-scale data processing.
- Designed and implemented **Delta Lake** architecture on **Databricks** to optimize data lake performance and reliability.
- Automated data workflows and job scheduling in **Databricks** using **Jobs API** and **Workflows**.
- Optimized **Spark jobs** in **Databricks** for performance, achieving 40% reduction in processing time.
- Implemented **Unity Catalog** for centralized data governance, enabling fine-grained access control and audit logging across **Databricks** workspaces.
- Managed data lineage and classification using **Unity Catalog** to enhance data security and compliance.
- Configured **Unity Catalog** with **Azure AD** groups to streamline role-based access across multiple environments.
- Deep experience with **AWS** services (**EMR**, **EC2**, **S3**, **Lambda**, **Redshift**, **RDS**, **CloudWatch**) and **Azure** services (**ADF**, **Data Lake Gen2**, **Synapse**, **Data Bricks**).
- Architected and implemented a unified **Data Lakehouse** solution using **Databricks Delta Lake**, enabling seamless integration of batch and streaming data, reducing infrastructure complexity by 40% and improving analytics delivery time from hours to minutes.
- Strong background in implementing real-time data **streaming** pipelines using **Kafka**, **Spark Streaming**, and **NoSQL** databases such as **HBase**, **Cassandra**, and **MongoDB**.
- Developed and maintained **ETL** frameworks using **PySpark**, **Python 3.11**, and **Talend** for structured and semi-structured data processing.
- Built interactive **dashboards** and **KPIs** using **Power BI 2022** and **Tableau 2024.1** to visualize performance metrics and business insights.
- Migrated on-premises **Hadoop** clusters and **Hive/MapReduce** workloads to **AWSEMR** and **Azure Synapse** for cost optimization and performance improvements.

- Experienced in integrating and automating pipelines using CI/CD tools such as **Jenkins, Git, Airflow,** and **Azure DevOps**.
- Data Governance & Compliance designed and implemented enterprise-wide data governance policies with Unity Catalog, Ranger, and Azure Purview, ensuring GDPR, HIPAA, SOC2 compliance.
- Built automated data lineage tracking for 100+ datasets using AWS Glue Data Catalog and Great Expectations, enabling full audit readiness.
- Knowledgeable in various file formats and data serialization techniques including **Avro, ORC, Parquet,** and **JSON**.
- Skilled in implementing data governance and security using **Kerberos, Sentry, Ranger,** and **SSL/TLS** for enterprise-grade compliance.
- Performance & Optimization Optimized Spark, SQL, and Delta Lake workloads using partition pruning, predicate pushdown, bucketing, and broadcast joins — reducing ETL runtime by 40–60%.
- Converted high-latency batch jobs to real-time streaming pipelines using Kafka + Spark Structured Streaming, reducing data availability time from hours to seconds.
- Extensive knowledge of data warehousing concepts and dimensional modeling for BI/reporting environments.
- Experienced in handling petabyte-scale data ingestion, transformation, and loading into **HDFS, S3,** and **Azure Blob Storage**.
- Responsible for developing data pipeline using **Spark, Scala,** Apache **Kafka** to ingest the data from CSV source and store in **HDFS** protected folder.
- Migrate databases to cloud platform **SQL Azure** and as well the performance tuning.
- Written **Map Reduce** code to process and parsing the data from various sources and storing parsed data into **HBase** and **Hive** using **HBase-Hive Integration**.
- Knowledge of **ETL** methods for data extraction, transformation and loading in corporate-wide **ETL** Solutions and **Data Warehouse** tools for reporting and data analysis.
- Extracted and updated the data into **HDFS** using **Sqoop** import and export.
- Created **Hive, Pig, SQL** and **HBase** tables to load large sets of structured, semi-structured and unstructured data coming from **UNIX, NoSQL** and a variety of portfolios.
- Hands on experience in installing configuring and using **Hadoop** ecosystem components like **Hadoop Map reduce HDFS HBase Hive Sqoop Pig Zookeeper** and **Flume**
- Experienced with the **Scala, Spark** improving the performance and optimization of the existing algorithms in **Hadoop** using **Spark Context, Spark -SQL, Pair RDD's, Spark YARN**

TECHNICAL SKILLS

Category	Technologies
Big Data & Processing	Hadoop (HDFS, MapReduce, YARN), Apache Spark (1.6 – 3.5), Spark SQL, Spark Streaming, Apache Flink, Pig, Hive (1.2 – 3.1), Sqoop, Flume, Impala, Oozie, Zookeeper, Tez
Cloud Platforms	AWS (EMR, EC2, S3, Lambda, Redshift, CloudWatch, CloudFormation, Lake Formation), Azure (Data Factory, Data Bricks, Synapse Analytics, Blob Storage, Data Lake Gen2, Cosmos DB, Azure DevOps)
Programming Languages	Python (2.7 – 3.11): Expert in ETL frameworks, data ingestion, automation (Pandas, PySpark, NumPy, SQLAlchemy) Scala (2.12): Spark applications, UDFs, streaming Java: MapReduce jobs, REST APIs, Spring Boot microservices
SQL & Databases	SQL (Oracle, MySQL, PostgreSQL, SQL Server): Query optimization, indexing, partitioning, stored procedures NoSQL: HBase, Cassandra, MongoDB, Delta Lake
Shell Scripting	Bash: Data loading automation, log parsing, pipeline orchestration on Linux environments
Other Programming Languages	C & C++: Algorithm design and optimization
JavaScript	Data visualization integration, building lightweight API layers for data access
Data Streaming & Messaging	Kafka (2.4 – 3.6), Azure Event Hub, Apache NiFi
Visualization & Reporting	Tableau (2024.1), Power BI (2015 – 2024), Kibana
CI/CD & DevOps Tools	Git, Jenkins, Azure DevOps, Docker, Airflow, Control-M
Serialization & Formats	JSON, Avro, Parquet, ORC, CSV
Security & Governance	Kerberos, Sentry, Ranger, SSL/TLS
Other Tools	Hue, Ambari, Jira, Confluence, Great Expectations, Kusto Explorer, GitHub

PROFESSIONAL EXPERIENCE

Client: Mayo Clinic, Rochester, MN

March 2023-Present

Role: Principal Big Data Engineer

Responsibilities:

- Designed and implemented data ingestion and transformation pipelines on **AWSEMR** using **Spark** 3.5 and **Python** 3.11 for real-time analytics.
- Built distributed data processing systems to ingest over 20 TB of data per day from **Kafka** topics and batch sources into **S3** and **Redshift**.
- Leveraged **AWS** Glue and **Lambda** functions to **orchestrate** and monitor **ETL** pipelines while optimizing for cost and performance.
- Created **Spark**-based batch jobs to cleanse, transform, and aggregate raw **JSON/CSV** data into curated **S3** layers.

- Developed data lake governance policies using **AWSLake Formation** and integrated with **Ranger** for fine-grained access control.
- Performed log analysis using **Elasticsearch** and **Kibana** to track and debug **streaming** job failures.
- Designed highly available and **scalable** solutions for real-time anomaly detection using **Spark Streaming** and Kafka.
- Built **Hive** external tables with partitions for efficient querying on large datasets stored in **S3**.
- Tuned **Spark** jobs using dynamic allocation, broadcast joins, and caching strategies for optimal performance.
- Automated cluster provisioning and deployment **scripts** using **AWSCloudFormation** and **Python**-based CLI tools.
- Led migration efforts from on-prem **Hadoop** to **AWSEMR**, reducing operational overhead by 30%.
- Configured **CloudWatch** metrics and custom dashboards to monitor job performance and cluster health.
- Defined data quality checks and alerts in **Airflow** using **Great Expectations** framework.
- Created **Tableau** dashboards to visualize **streaming** pipeline metrics and alerting trends.
- Mentored a team of 5 junior data engineers and conducted weekly knowledge-sharing sessions on **AWS** best practices.
- Installed and configured **Hive, Pig, Sqoop, Flume** and **Oozie** on the **Hadoop** cluster.
- Developed a strategy for Full load and incremental load using **Sqoop**.
- Documented the requirements including the available code which should be implemented using **Spark, Hive, HDFS, HBase** and **ElasticSearch**.
- Developed a **Python** Script to load the **CSV** files into the **S3** buckets and created **AWSS3** buckets, performed folder management in each bucket, managed logs and objects within each bucket.
- Good knowledge in Cluster coordination services through **Zookeeper** and **Kafka**.
- Experienced in writing live Real-time Processing using **Spark Streaming** with **Kafka**
- **Assisted** in Cluster maintenance, cluster monitoring, adding and removing cluster nodes and installed and configured **Hadoop, Map Reduce, HDFS**, developed multiple **Map Reduce** jobs in **java** for data cleaning and pre-processing.
- Involved in file movements between **HDFS** and **AWS S3** and extensively worked with **S3** bucket in **AWS**.
- Created data partitions on large data sets in **S3** and DDL on partitioned data.
- Converted all **Hadoop** jobs to run in **EMR** by configuring the cluster according to the data size.

Environment: AWS (EMR, S3, EC2, Lambda, Redshift, CloudFormation, CloudWatch), Spark 3.5, Python 3.11, Hive 3.1, Kafka 3.6, Ranger, Tableau 2024.1, Airflow, Elasticsearch, Kibana, Git, Jenkins

Client: Barclays Investment Bank. Whippany, NJ

April 2021 - February 2023

Role: Lead Big Data Engineer

Responsibilities:

- Developed ingestion processes using **Kafka** and **Azure Event Hub** to bring **streaming** data into **Data Lake Gen2**.
- Designed **ETL** pipelines to clean and prepare structured/unstructured data from **REST APIs** and IoT feeds.
- Implemented **Spark** jobs using **PySpark 3.9** and **Spark SQL** to transform raw data and store curated datasets in **Parquet** format.
- Enabled role-based access control across **Azure** services using **Azure AD** and service principals.
- Deployed and maintained CI/CD pipelines for deploying **PySpark scripts** and ADF artifacts via **Azure DevOps**.
- Scheduled and **orchestrated** jobs using Apache **Airflow** for data integration between **Blob Storage, Synapse**, and **SQL** pools.

- Built real-time fraud detection engine using **Spark** Structured **Streaming** and **Azure Cosmos DB**.
- Tuned **Spark** configurations for memory management and executor optimization, resulting in a 40% speedup.
- Defined reusable data quality rules and validation logic in **Data Factory** for ingestion-level checks.
- Conducted POC on Kusto Query Language (KQL) and **Azure** Data Explorer for telemetry analytics.
- Collaborated with product teams to understand analytics needs and translate into robust, **scalable** solutions.
- Worked with business analysts to define KPIs and developed **Power BI** 2022 dashboards for reporting.
- Performed metadata management using Purview and integrated with **Data Factory** lineage view.
- Provided L3 support and handled on-call issues related to **Spark** job failures and ADF latency.
- Initiated cloud cost optimization by introducing spot instances and auto-scaling policies in **Azure** Databricks.
- Implementing various resources in **Azure** using **Azure** Portal, PowerShell on ARM deployment models.
- Designed and implemented data ingestion **Spark streaming** framework from various data source like **REST API**, **Kafka** using **Spark StreamingScalaAPI** and **Kafka**.
- Wrote, compiled, and executed programs as necessary using **Apache Spark** in **Scala** to perform **ETL** jobs with ingested data.
- Collaborated with data teams to showcase the projects KPIs by using Big Data System **Azure** Data Lake, Scope and **Azure** data Explorer (Kusto).
- Using **Azure SynapseSQL** Pool to load in Fact and/or Dimension Tables.

Environment: Azure (Data Factory, Synapse, Blob Storage, Data Bricks, Data Lake Gen2, Azure DevOps), Kafka, PySpark 3.9, Delta Lake, Power BI 2022, Kusto Explorer, Cosmos DB, Git, Airflow, REST API, KQL

Client: Macy's Duluth, GA

September 2019 -March 2021

Role: Senior Big Data Engineer

Responsibilities:

- Created end-to-end data pipeline frameworks using **Spark** 2.4 with **Scala** and **Hive** 2.3 for processing clickstream and transaction data.
- Ingested multi-terabyte datasets from **Oracle** and **MySQL** using **Sqoop** and transformed using **MapReduce**.
- Integrated **HBase** and **Hive** for storage and queried using **Hive** QL to perform behavioral analysis.
- Developed batch processing jobs to clean, join, enrich, and load data into **HDFS** and **Hive** tables.
- Used **Oozie** to coordinate and schedule **Hadoop** jobs across different time zones.
- Implemented **Kafka** consumers using **Scala** for high-throughput ingestion of application logs.
- Tuned and optimized **MapReduce** jobs to improve performance by 35% over legacy implementation.
- Built dashboards using **Power BI** to monitor performance of **Hive** queries and cluster utilization.
- Leveraged **Avro** and **Parquet** for storage format optimization and schema evolution.
- Conducted unit testing and performance benchmarking for various ingestion strategies.
- Handled 24/7 production support for **Spark** and **MapReduce** jobs with on-call rotation.
- Collaborated with QA and data analysts for UAT and sign-off before production release.
- Designed reusable **Hive** UDFs for custom transformations and text parsing.
- Developed monitoring **scripts** in **Python** and **Shell** to validate job completion.
- Responsible for developing data pipeline with **Amazon AWS** to extract the data from weblogs and store in **HDFS** and worked extensively with **Sqoop** for importing metadata from **Oracle**.
- Analyzed the data by performing **Hive** queries (**HiveQL**) and running **Pig scripts** (**Pig Latin**) for data ingestion

- Exporting of a result set from **HIVE** to **MySQL** using **Sqoop** export tool for further processing.
- Installed and configured **Hive, Pig, Sqoop, Flume** and **Oozie** on the **Hadoop** cluster.
- Involved in support for **Amazon AWS and RDS** to host static/media files and the database into **Amazon Cloud**.
- Implemented the automated workflows for all the jobs using the **Oozie** and **shell** script.
- Used **Spark SQL** functions to move data from stage **hive** tables to fact and dimension tables.
- Created Micro services using **AWS Lambda and API Gateway** using **REST API**.
- Generated report on predictive analytics using **Python** and **Tableau** including visualizing model performance and prediction results.
- Implemented data ingestion and handling clusters in real time processing using **Kafka**.

Environment: AWS (EMR, S3, EC2), Hive 2.3, HDFS, Sqoop, Spark 2.4, Scala 2.12, MapReduce, Oozie, Power BI, Kafka 2.4, Avro, Parquet, Git, Shell

Client: Comcast Philadelphia, PA

January 2017 - August 2019

Role: Mid-Level Big Data Engineer

Responsibilities:

- Developed **scalable** batch **ETL** jobs using **Spark 2.1** and **Hive 2.1** to process raw data into consumable datasets.
- Built and maintained ingestion pipelines using **Flume** and **Sqoop** for log and metadata ingestion.
- Created **Hive** external tables with custom SerDes to handle semi-structured **JSON** data.
- Used **Pig** and **Hive** for preprocessing and aggregating log events for user segmentation.
- **Assisted** in implementing **S3**-based data lake and used **AWS Glue** for data cataloging.
- Collaborated with DevOps team to containerize **Spark** jobs using **Docker** and deploy via **Jenkins**.
- Designed partitioning strategies to enhance **Hive** query performance over large datasets.
- Troubleshooted long-running jobs by analyzing logs using **YARN** Resource Manager and **Hue**.
- Implemented data deduplication logic using **Spark** Window functions.
- Performed daily health checks of **Hadoop** cluster and managed job failures.
- Automated metadata refresh and **Hive** schema reconciliation **scripts** in **Python 3.6**.
- Created and maintained version-controlled codebase in **Git** with proper branching.
- Participated in weekly sprint meetings to plan and review deliverables.
- Engaged in code reviews and peer testing for collaborative development.
- Conduct systems design, feasibility and cost studies and recommend cost-effective cloud solutions such as **Amazon Web Services (AWS)**.
- Created Session Beans and controller Servlets for handling **HTTP** requests from **Talend**
- Utilized Waterfall methodology for team and project management.
- Used **Git** for version control with Data Engineer team and Data Scientists colleagues.

Environment: AWS (EMR, S3), Spark 2.1, Hive 2.1, Pig, Sqoop, Flume, Docker, Python 3.6, Jenkins, Git, Hue, YARN, Shell

Client: RedPine Solutions, Hyderabad, India

June 2013 - September 2016

Role: Junior Data Engineer

Responsibilities:

- Supported **data engineering** team in writing **Hive QLscripts** and optimizing queries.
- Maintained and ingested small data batches into **HDFS** using **Flume** and **Sqoop**.
- Conducted data validation checks for consistency and completeness in **HDFS**.
- Wrote simple **Pig** Latin **scripts** to perform transformation on structured log files.
- Documented **Hive** tables, partitions, and lineage for reporting.

- Generated summary statistics using **Spark RDDs** and basic transformations in **Scala**.
- Created **shellscripts** for automating daily data loads from **UNIX** servers to **HDFS**.
- **Assisted** in creating internal dashboards using **Tableau** for operational KPIs.
- Implemented access controls on **Hive** tables using **Ranger** policies.
- Gained familiarity with **Kerberos** authentication and job scheduling via **Oozie**.
- Built unit tests for data quality checks using **Python** and **Hive** queries.
- Participated in **Hadoop** cluster upgrades and testing under supervision.
- Contributed to team knowledge base on **Spark** job tuning techniques.
- Performed backup and archival of **Hive** metadata and data lineage documentation.
- Supported business users in **ad-hoc** reporting using **Hive** queries.

Environment: Hadoop 2.7, HDFS, Hive 1.2, Sqoop, Pig, Spark 1.6, Scala, Tableau, Python 2.7, Shell, Ranger, Oozie